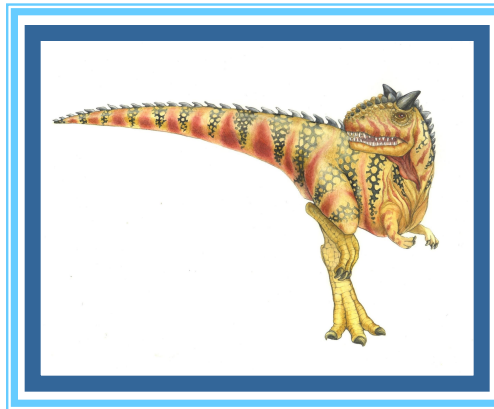


Chapter 6: Synchronization Tools





Outline

- Background
- The Critical-Section Problem
- Peterson's Solution
- Hardware Support for Synchronization
- Mutex Locks
- Semaphores
- Monitors
- Liveness
- Evaluation





Objectives

- Describe the critical-section problem and illustrate a race condition
- Illustrate hardware solutions to the critical-section problem using memory barriers, compare-and-swap operations, and atomic variables
- Demonstrate how mutex locks, semaphores, monitors, and condition variables can be used to solve the critical section problem
- Evaluate tools that solve the critical-section problem in low-, Moderate-, and high-contention scenarios





Background

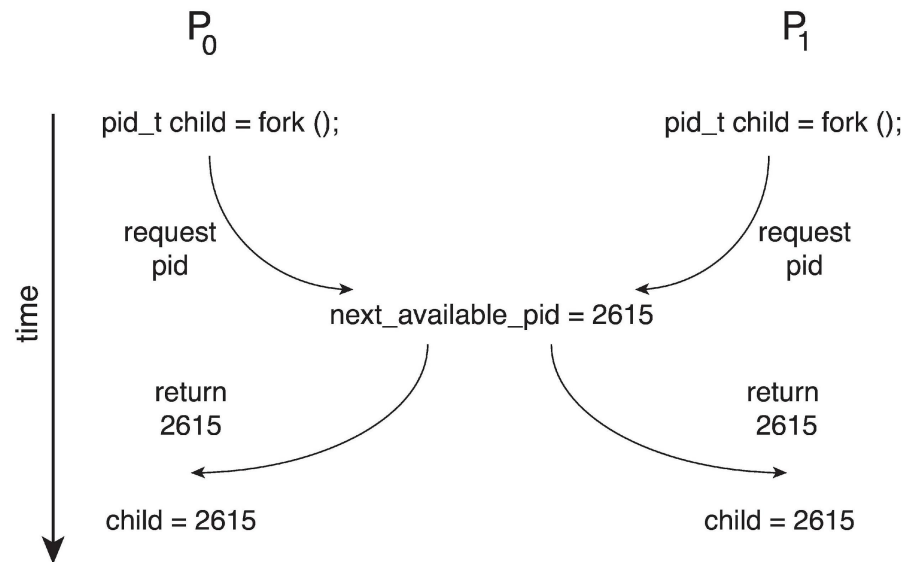
- Processes can execute concurrently
 - May be interrupted at any time, partially completing execution
- Concurrent access to shared data may result in data inconsistency
- Maintaining data consistency requires mechanisms to ensure the orderly execution of cooperating processes
- We illustrated in chapter 4 the problem when we considered the Bounded Buffer problem with use of a counter that is updated concurrently by the producer and consumer,. Which lead to race condition.





Race Condition

- Processes P_0 and P_1 are creating child processes using the `fork()` system call
- Race condition on kernel variable `next_available_pid` which represents the next available process identifier (pid)



- Unless there is a mechanism to prevent P_0 and P_1 from accessing the variable `next_available_pid` the same pid could be assigned to two different processes!





Critical Section Problem

- Consider system of n processes $\{p_0, p_1, \dots, p_{n-1}\}$
- Each process has **critical section** segment of code
 - Process may be changing common variables, updating table, writing file, etc.
 - When one process in critical section, no other may be in its critical section
- **Critical section problem** is to design protocol to solve this
- Each process must ask permission to enter critical section in **entry section**, may follow critical section with **exit section**, then **remainder section**





Critical Section

- General structure of process P_i

```
while (true) {  
    entry section  
    critical section  
    exit section  
    remainder section  
}
```





Critical-Section Problem (Cont.)

Requirements for solution to critical-section problem

1. **Mutual Exclusion** - If process P_i is executing in its critical section, then no other processes can be executing in their critical sections
2. **Progress** - If no process is executing in its critical section and there exist some processes that wish to enter their critical section, then the selection of the process that will enter the critical section next cannot be postponed indefinitely
3. **Bounded Waiting** - A bound must exist on the number of times that other processes are allowed to enter their critical sections after a process has made a request to enter its critical section and before that request is granted
 - Assume that each process executes at a nonzero speed
 - No assumption concerning **relative speed** of the n processes





Interrupt-based Solution

- Entry section: disable interrupts
- Exit section: enable interrupts
- Will this solve the problem?
 - What if the critical section is code that runs for an hour?
 - Can some processes starve – never enter their critical section.
 - What if there are two CPUs?





Software Solution 1

- Two process solution
- Assume that the **load** and **store** machine-language instructions are atomic; that is, cannot be interrupted
- The two processes share one variable:
 - **int turn;**
- The variable **turn** indicates whose turn it is to enter the critical section
- initially, the value of **turn** is set to *i*





Algorithm for Process P_i

```
while (true) {  
  
    while (turn == j);  
  
    /* critical section */  
    turn = j;  
  
    /* remainder section */  
  
}
```





Correctness of the Software Solution

- Mutual exclusion is preserved

P_i enters critical section only if:

turn = i

and **turn** cannot be both 0 and 1 at the same time

- What about the Progress requirement?
- What about the Bounded-waiting requirement?





Peterson's Solution

- Two process solution
- Assume that the **load** and **store** machine-language instructions are atomic; that is, cannot be interrupted
- The two processes share two variables:
 - **int** `turn`;
 - **boolean** `flag[2]`
- The variable **turn** indicates whose turn it is to enter the critical section
- The **flag** array is used to indicate if a process is ready to enter the critical section.
 - **flag[i] = true** implies that process P_i is ready!





Algorithm for Process P_i

```
while (true){  
  
    flag[i] = true;  
    turn = j;  
    while (flag[j] && turn == j)  
        ;  
  
    /* critical section */  
  
    flag[i] = false;  
  
    /* remainder section */  
  
}
```





Correctness of Peterson's Solution

- Provable that the three CS requirement are met:
 1. Mutual exclusion is preserved

P_i enters CS only if:
either **flag[j] = false** or **turn = i**
 2. Progress requirement is satisfied
 3. Bounded-waiting requirement is met





Peterson's Solution and Modern Architecture

- Although useful for demonstrating an algorithm, Peterson's Solution is not guaranteed to work on modern architectures.
 - To improve performance, processors and/or compilers may reorder operations that have no dependencies
- Understanding why it will not work is useful for better understanding race conditions.
- For single-threaded this is ok as the result will always be the same.
- For multithreaded the reordering may produce inconsistent or unexpected results!





Synchronization Hardware

- Many systems provide hardware support for implementing the critical section code.
- Uniprocessors – could disable interrupts
 - Currently running code would execute without preemption
 - Generally too inefficient on multiprocessor systems
 - 4 Operating systems using this not broadly scalable
- We will look at three forms of hardware support:
 1. Hardware instructions
 2. Atomic variables





Hardware Instructions

- Special hardware instructions that allow us to either *test-and-modify* the content of a word, or to *swap* the contents of two words atomically (uninterruptedly.)
 - **Test-and-Set** instruction
 - **Compare-and-Swap** instruction





The test_and_set Instruction

- Definition

```
boolean test_and_set (boolean
*target)
{
    boolean rv = *target;
    *target = true;
    return rv;
}
```

- Properties

- Executed atomically
- Returns the original value of passed parameter
- Set the new value of passed parameter to **true**





Solution Using test_and_set()

- Shared boolean variable `lock`, initialized to `false`
- Solution:

```
do {  
    while (test_and_set(&lock))  
        ; /* do nothing */  
  
    /* critical section */  
  
    lock = false;  
    /* remainder section */  
} while (true);
```

- Does it solve the critical-section problem?





The compare_and_swap Instruction

■ Definition

```
int compare_and_swap(int *value, int expected, int new_value)
{
    int temp = *value;
    if (*value == expected)
        *value = new_value;
    return temp;
}
```

■ Properties

- Executed atomically
- Returns the original value of passed parameter **value**
- Set the variable **value** the value of the passed parameter **new_value** but only if ***value == expected** is true. That is, the swap takes place only under this condition.





Solution using compare_and_swap

- Shared integer `lock` initialized to 0;
- Solution:

```
while (true){  
    while (compare_and_swap(&lock, 0, 1) != 0)  
        ; /* do nothing */  
  
    /* critical section */  
  
    lock = 0;  
  
    /* remainder section */  
}
```

- Does it solve the critical-section problem?





Mutex Locks

- Previous solutions are complicated and generally inaccessible to application programmers
- OS designers build software tools to solve critical section problem
- Simplest is mutex lock
 - Boolean variable indicating if lock is available or not
- Protect a critical section by
 - First **acquire()** a lock
 - Then **release()** the lock
- Calls to **acquire()** and **release()** must be **atomic**
 - Usually implemented via hardware atomic instructions such as compare-and-swap.
- But this solution requires **busy waiting**
 - This lock therefore called a **spinlock**





Solution to CS Problem Using Mutex Locks

```
while (true) {  
    acquire lock  
  
    critical section  
  
    release lock  
  
    remainder section  
}
```





Semaphore

- Synchronization tool that provides more sophisticated ways (than Mutex locks) for processes to synchronize their activities.
- Semaphore **S** – integer variable
- Can only be accessed via two indivisible (atomic) operations

- **wait()** and **signal()**

4 Originally called **P()** and **V()**

- Definition of the **wait()** operation

```
wait(S) {  
    while (S <= 0)  
        ; // busy wait  
    S--;  
}
```

- Definition of the **signal()** operation

```
signal(S) {  
    S++;  
}
```





Semaphore (Cont.)

- **Counting semaphore** – integer value can range over an unrestricted domain
- **Binary semaphore** – integer value can range only between 0 and 1
 - Same as a **mutex lock**
- Can implement a counting semaphore **S** as a binary semaphore
- With semaphores we can solve various synchronization problems





Semaphore Usage Example

- Solution to the CS Problem

- Create a semaphore “**mutex**” initialized to 1

```
wait(mutex) ;
```

```
CS
```

```
signal(mutex) ;
```

- Consider P_1 and P_2 that with two statements S_1 and S_2 and the requirement that S_1 to happen before S_2

- Create a semaphore “**synch**” initialized to 0

```
P1 :
```

```
S1 ;
```

```
signal(synch) ;
```

```
P2 :
```

```
wait(synch) ;
```

```
S2 ;
```





Semaphore Implementation

- Must guarantee that no two processes can execute the **wait()** and **signal()** on the same semaphore at the same time
- Thus, the implementation becomes the critical section problem where the **wait** and **signal** code are placed in the critical section
- Could now have **busy waiting** in critical section implementation
 - But implementation code is short
 - Little busy waiting if critical section rarely occupied
- Note that applications may spend lots of time in critical sections and therefore this is not a good solution





Semaphore Implementation with no Busy waiting

- With each semaphore there is an associated waiting queue
- Each entry in a waiting queue has two data items:
 - Value (of type integer)
 - Pointer to next record in the list
- Two operations:
 - **block** – place the process invoking the operation on the appropriate waiting queue
 - **wakeup** – remove one of processes in the waiting queue and place it in the ready queue





Implementation with no Busy waiting (Cont.)

- Waiting queue

```
typedef struct {  
    int value;  
    struct process *list;  
} semaphore;
```





Implementation with no Busy waiting (Cont.)

```
wait(semaphore *S) {
    S->value--;
    if (S->value < 0) {
        add this process to S->list;
        block();
    }
}

signal(semaphore *S) {
    S->value++;
    if (S->value <= 0) {
        remove a process P from S->list;
        wakeup(P);
    }
}
```





Problems with Semaphores

- Incorrect use of semaphore operations:
 - `signal(mutex) ... wait(mutex)`
 - `wait(mutex) ... wait(mutex)`
 - Omitting of `wait(mutex)` and/or `signal(mutex)`
- These – and others – are examples of what can occur when semaphores and other synchronization tools are used incorrectly.





Single Resource allocation

- Allocate a single resource among competing processes using priority numbers that specifies the maximum time a process plans to use the resource

```
R.acquire(t) ;  
    ...  
    access the resource;  
    ...  
  
R.release;
```

- Where R is an instance of type **ResourceAllocator**





Single Resource allocation

- Allocate a single resource among competing processes using priority numbers that specifies the maximum time a process plans to use the resource
- The process with the shortest time is allocated the resource first
- Let R is an instance of type **ResourceAllocator** (next slide)
- Access to **ResourceAllocator** is done via:

```
R.acquire(t) ;  
    ...  
    access the resource;  
    ...  
R.release;
```

- Where **t** is the maximum time a process plans to use the resource



End of Chapter 6

